

Outset · AI in Healthcare · Day 3 · Group 5

Predicting heart disease from a checkup

A tabular model, and the fairness question at its center: does it work for women?

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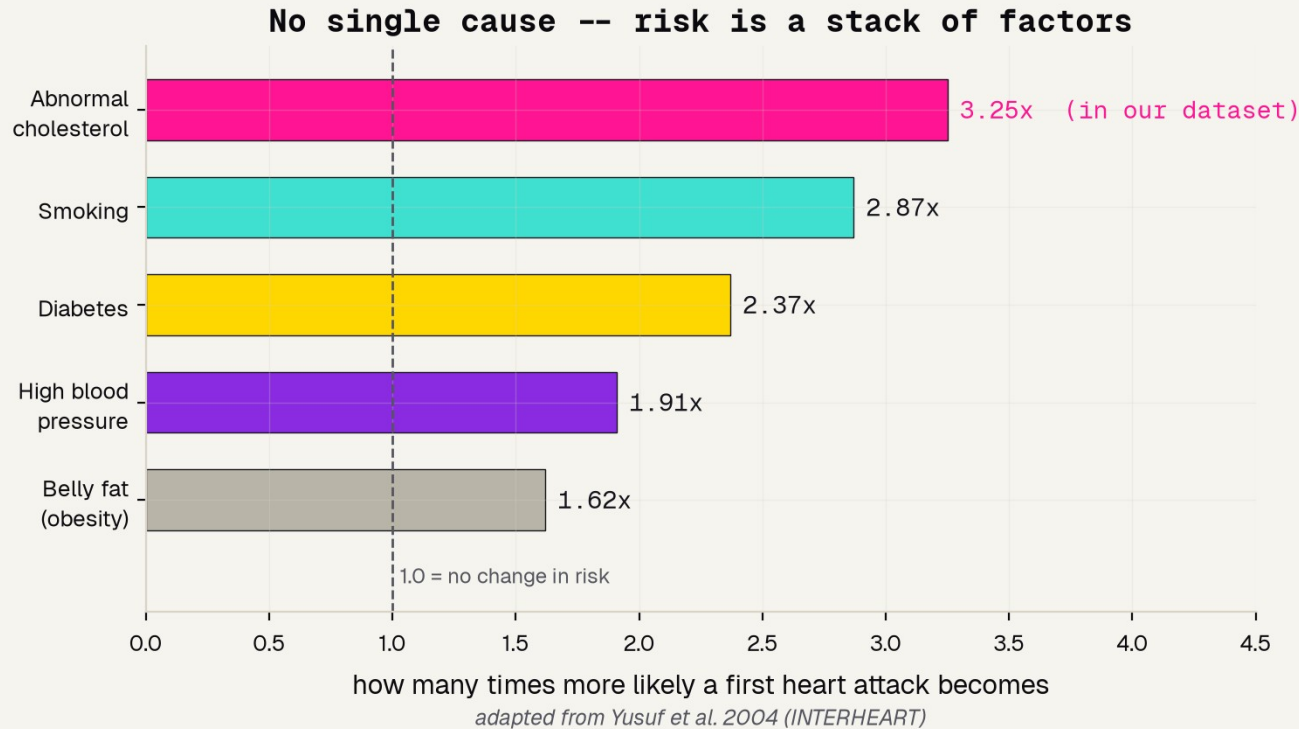
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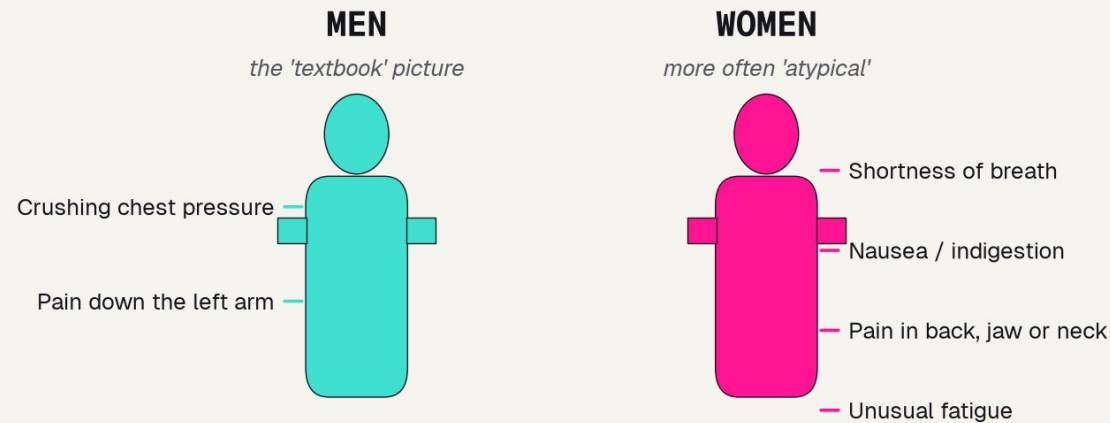
BACKGROUND

How doctors actually estimate heart risk



- No doctor diagnoses heart disease from one number
- They combine several routine checkup measurements into a single risk estimate, the way the Framingham Risk Score does. Worldwide, about nine risk factors led by cholesterol and smoking explain over 90% of first heart attacks. Risk is a stack of factors, not a single cause.

Women's heart disease is under-diagnosed



The 'classic' heart-attack symptoms are the MALE ones -- one reason women's heart attacks are more often missed.

concept from an AHA scientific statement (Mehta et al. 2016) and van Oosterhout et al. 2020

- Here is the problem that drives our project
- The “textbook” heart attack, crushing chest pain down the left arm, is the male picture. Women more often feel back pain, nausea, and shortness of breath, so their heart attacks get missed. The American Heart Association documents that women are misdiagnosed and under-treated more than men.

METHODS

The task and the data

303 patients

The Cleveland Heart Disease dataset, from a 1980s study.

13 features

Age, sex, blood pressure, cholesterol, chest-pain type, exercise-ECG results.

Predict disease

1 = has heart disease, 0 = does not. Graded on held-out patients.

Four models, one honest test

Logistic Regression

Simple, explainable, and here the best of the four.

Random Forest & CatBoost

Tree ensembles, the workhorses of tabular data.

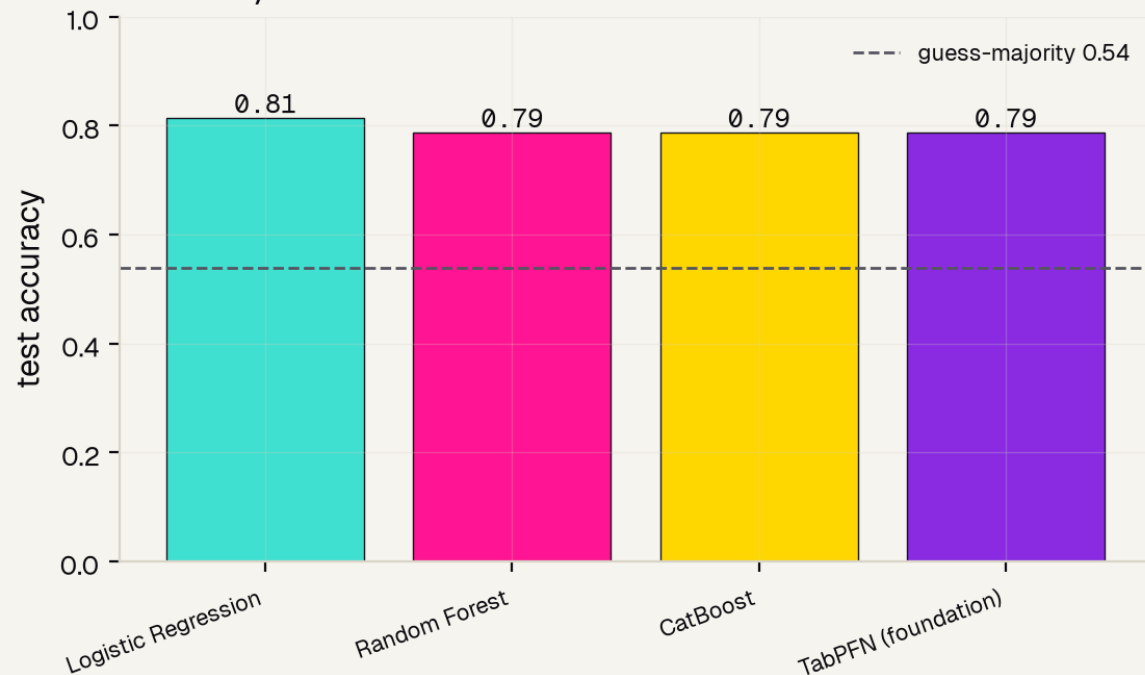
TabPFN

A foundation model for tables, like the one from Day 2.

RESULTS

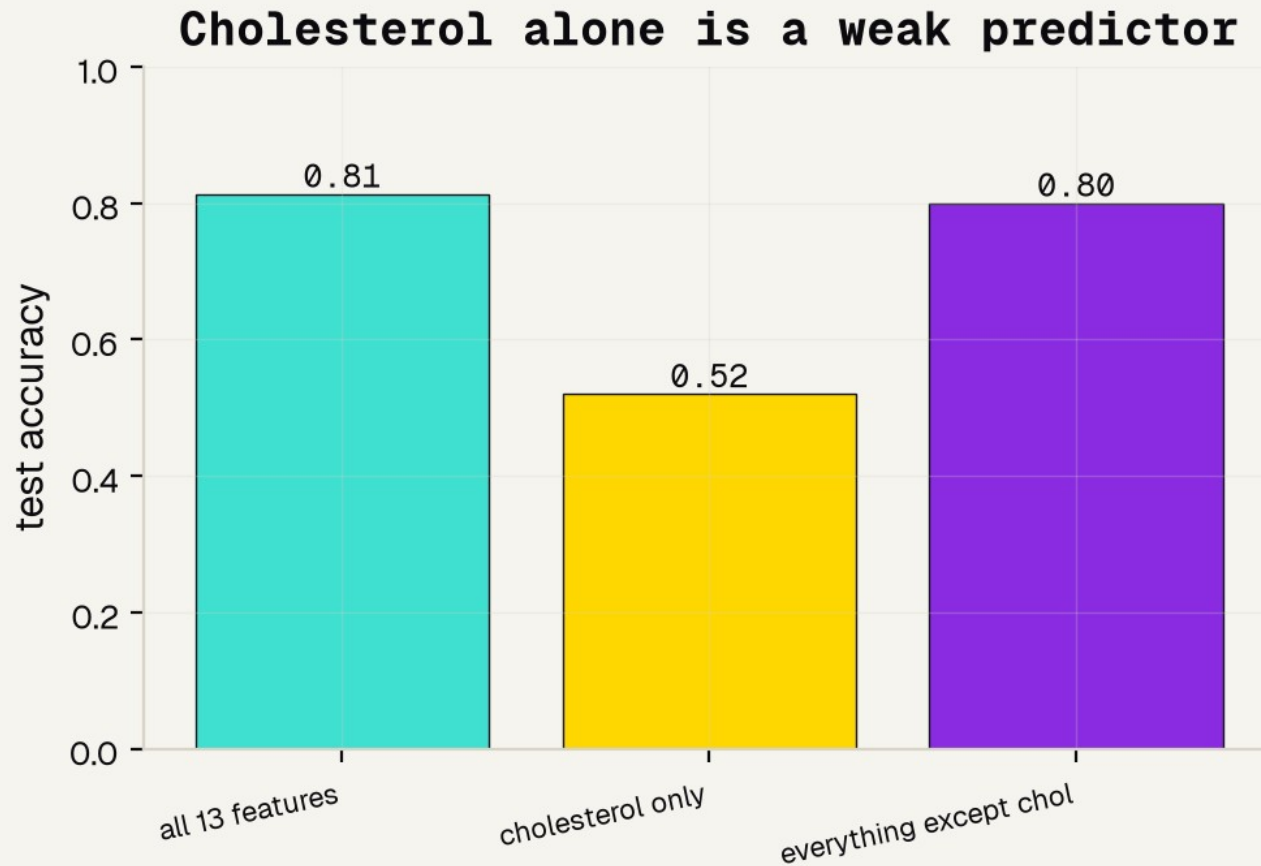
The data sets the ceiling

Four models, similar scores -- the data sets the ceiling



- All four models land between 0.79 and 0.81, comfortably above the coin-flip baseline, and they basically tie each other. That is the lesson: on clean tabular data the model barely matters. Swapping a plain logistic regression for a fancy foundation model buys almost nothing. Three hundred patients from the 1980s contain only so much signal, and no model can invent more.

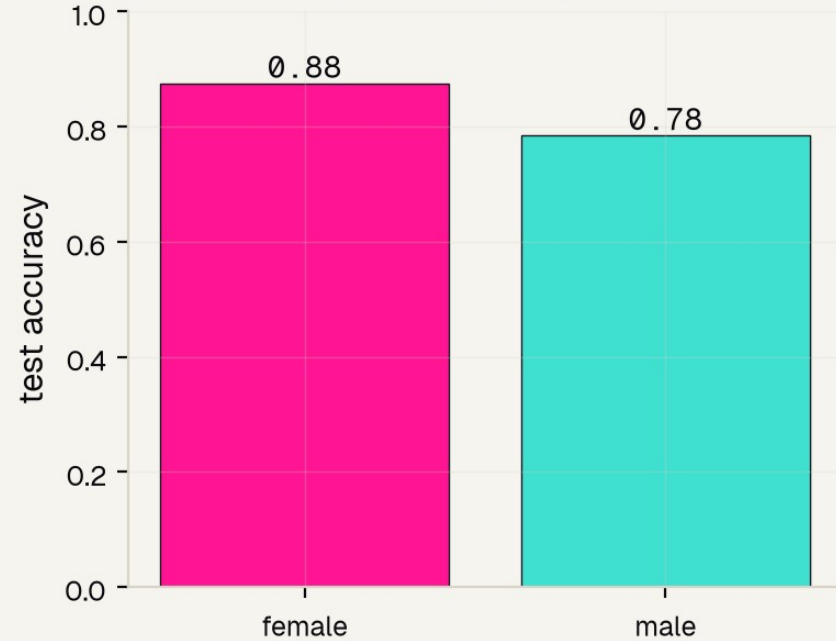
Cholesterol alone is a weak predictor



- Everyone “knows” cholesterol causes heart disease, so we gave a model only the cholesterol column. It scored about 0.52, basically a coin flip. Meanwhile the checkup with cholesterol removed scored about 0.80. Cholesterol is a major cause of heart disease over decades, but on its own it is a poor clue for who has disease right now. That is why doctors combine many measurements.

Accurate overall, but not equally by sex

Accurate overall -- but not equally for women and men



- This is the point of the project
- One accuracy number can hide unfairness. When we grade the model separately for women and men, it is about 0.88 accurate for women and 0.78 for men, a gap of about 0.09. An accuracy gap in either direction means the model behaves differently depending on your sex. You only see it because you looked.



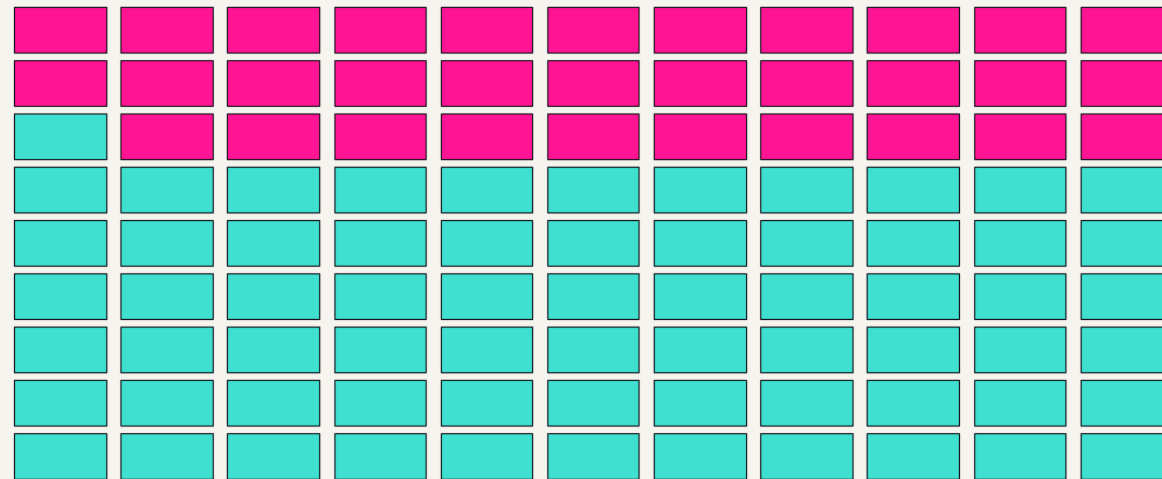
Validation

BEING HONEST

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The data is small and male-skewed

Who is in the data? Each square = 3 patients



men -- 201 patients (about 2 of every 3) women -- 96

from the Cleveland Heart Disease data (Detrano et al. 1989)

- A model can only learn from who is in its data, and ours is about 2:1 male
- With far fewer women, the model gets less practice on them, and the way patients were selected in the 1980s made the male rows sicker. That skew is not a bug we can fix with a better model, it is a property of the data, and it is exactly the bias that makes heart disease harder to catch in women.

What this project can and cannot say

It can

Show that checkup features carry real signal, that no single number is the whole story, and that accuracy can be unequal across sex.

It cannot

Say anything trustworthy about today's patients. This is 303 people from one 1980s study.

Watch the metric

Missing a real heart attack is far worse than a false alarm. A deployed tool would be tuned for sensitivity, not raw accuracy.

REFERENCES

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Sources

Cardiac risk scoring

Wilson 1998, Framingham Risk Score, Circulation. Yusuf 2004, the INTERHEART study, Lancet.

Under-diagnosis in women

Mehta 2016, AHA scientific statement, Circulation. van Oosterhout 2020, JAHA. Bugiardini and Bairey Merz 2005, JAMA.

The dataset

Detrano 1989, American Journal of Cardiology. UCI Machine Learning Repository, Heart Disease Data Set.

Thank You

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夢想